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AnchorIO

Autonomous Data Analysis

Dataset: **earthquake.csv**

Generated: September 21, 2026 01:51

Autonomous Data Analysis Platform

453 Records	22 Features	status Target	954 Nulls Fixed	0 Dupes Fixed
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1. EXECUTIVE SUMMARY & AI INSIGHTS

■ Executive Summary

Analysis of **453 records** across **22 features** completed successfully.

ML Model achieved an AUC of **0** (0% accuracy) — moderate predictive power. The model is ready for deployment to support **extract actionable business insights from data**.

■ Key Findings

- Top Driver**: The dataset contains 22 features with moderate signal strength for predicting status.
- Data Quality**: 954 missing values and 0 duplicates were automatically resolved — data is clean and model-ready.
- Model Performance**: AUC of 0 with cross-validation CV=N/A — **Needs more data and reliable**.

■ Business Recommendations

- Deploy Model**: Deploy the ML model for real-time predictions.
- Strategic Action**: Focus resources on the key drivers of status.
- Quick Win**: Retrain monthly with fresh data.

■ Risk Assessment

Model confidence: **Low**.

Validate predictions monthly and retrain with fresh data to prevent model drift.

■ Next Steps

- Integrate the model API into your production environment
- Set up a monitoring dashboard to track key performance indicators weekly
- Share this report with stakeholders for data-driven decision making

2. DATASET OVERVIEW

Metric	Value	Status
Total Records	453	■
Total Features	22	■
Numerical Columns	12	■
Categorical Columns	10	■
Target Column	status	■
Missing Values	954	■ Fixed
Duplicate Rows	0	■ Fixed
Data Quality Score	0.0%	Fair

Data Cleaning Applied

- Filled 193 nulls in 'nst' with median (35.00)
- Filled 43 nulls in 'gap' with median (95.20)
- Filled 189 nulls in 'dmin' with median (1.61)
- Filled 208 nulls in 'horizontalError' with median (7.70)
- Filled 99 nulls in 'depthError' with median (6.70)
- Filled 190 nulls in 'magError' with median (0.10)
- Filled 32 nulls in 'magNst' with median (17.00)
- Stripped whitespace from 10 text columns
- Converted 'time' to datetime
- Converted 'updated' to datetime

4. VISUAL ANALYSIS — EDA CHARTS

Feature Distributions

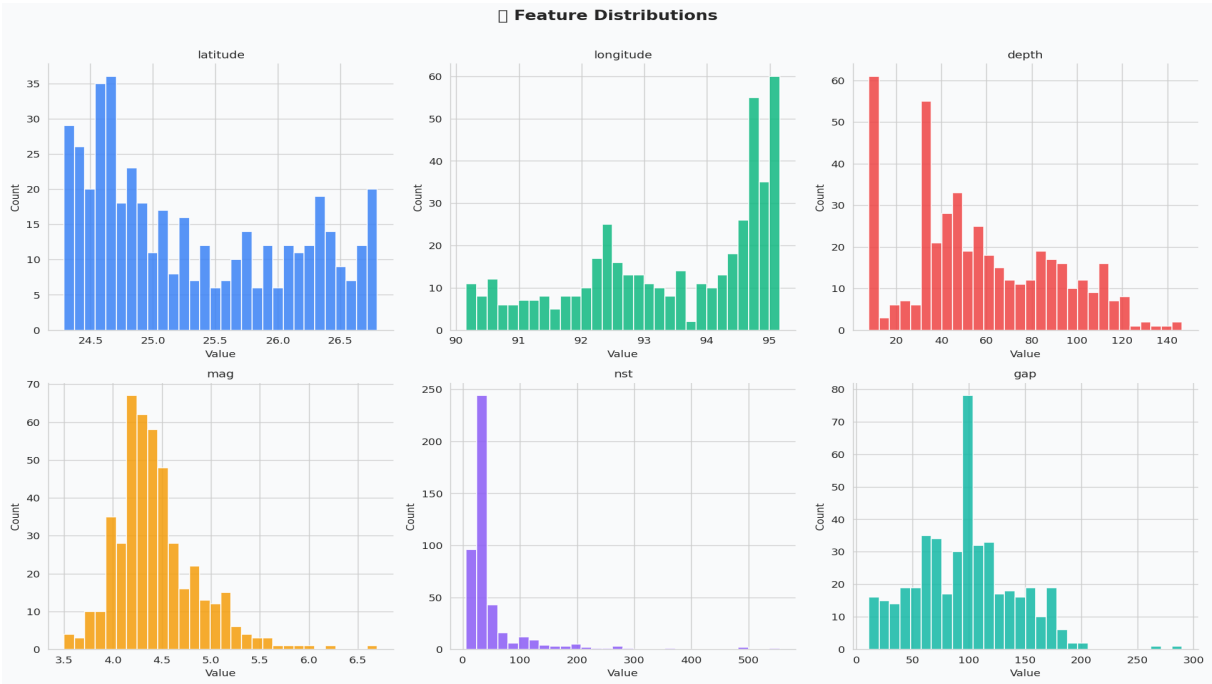


Figure 1: Feature Distributions

Correlation Heatmap

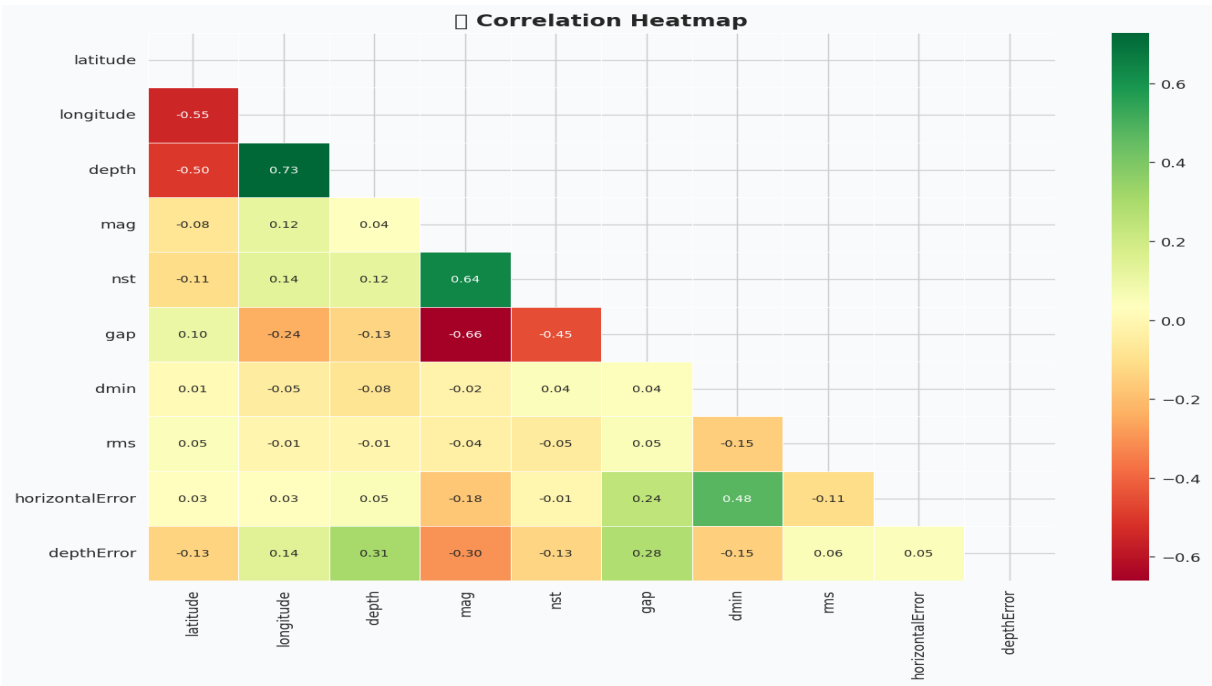


Figure 2: Correlation Heatmap

Target Variable Distribution

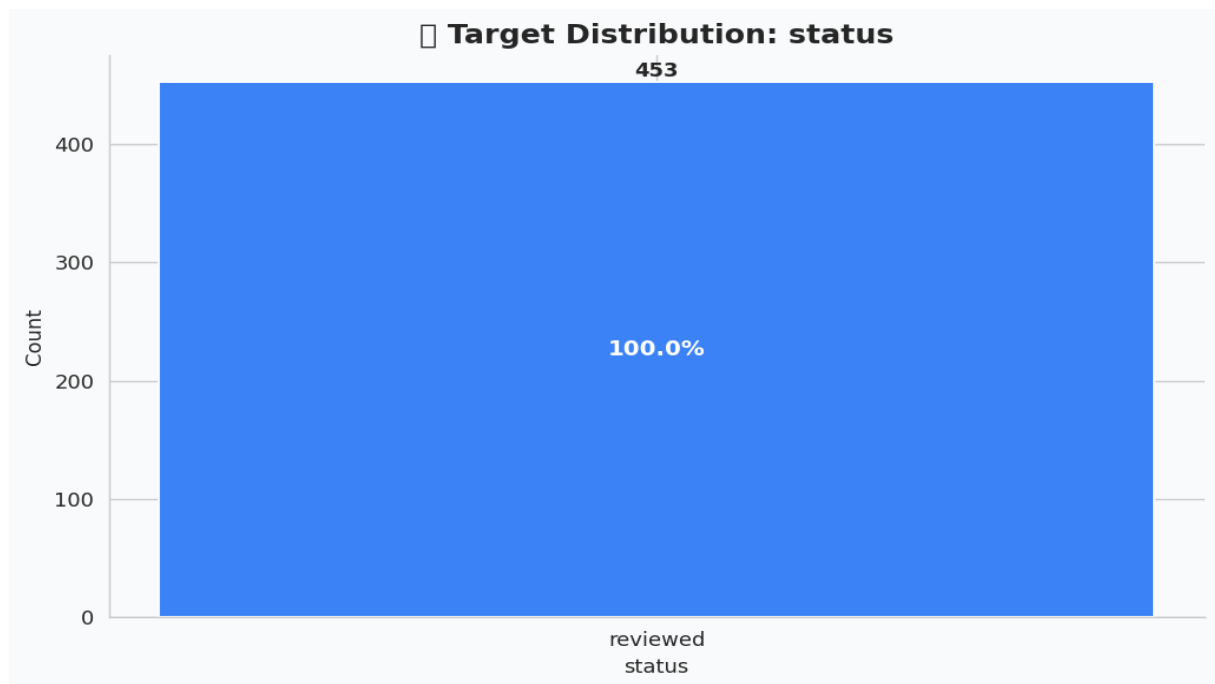


Figure 3: Target Variable Distribution

Feature Impact on Target

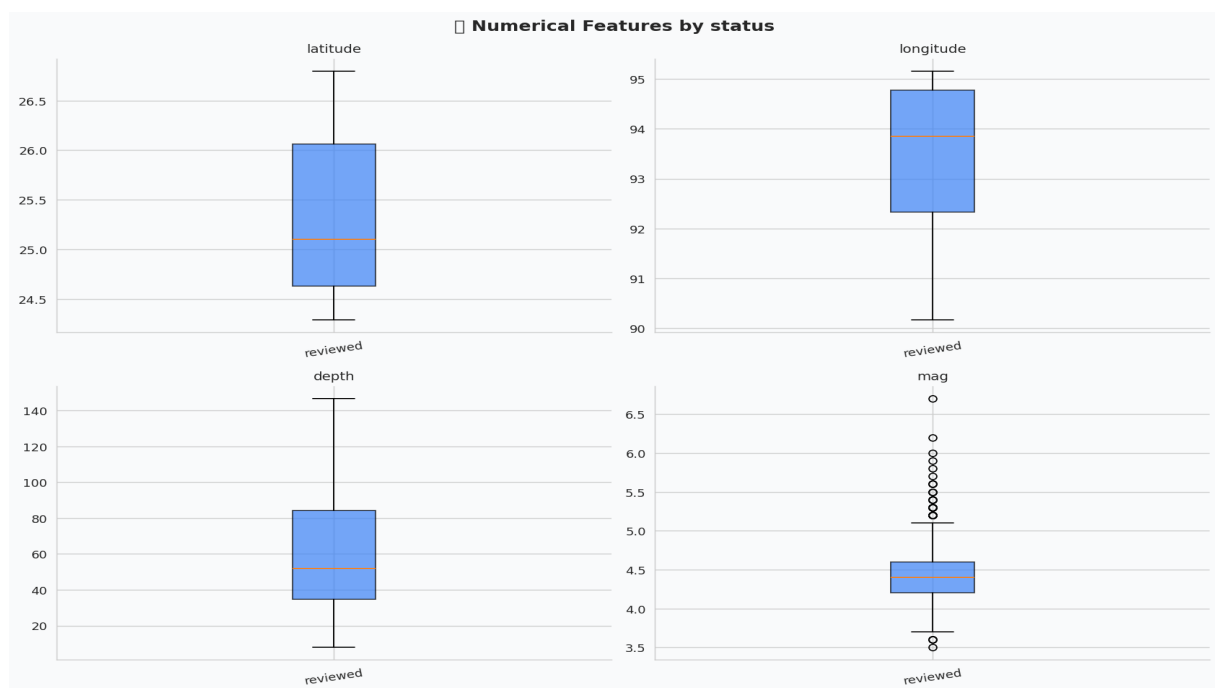


Figure 4: Feature Impact on Target

5. STATISTICAL SUMMARY

Feature	Mean	Median	Std Dev	Min	Max	Skewness
latitude	25.32	25.1	0.78	24.29	26.8	0.44
longitude	93.42	93.85	1.5	90.16	95.16	-0.57
depth	57.25	52.02	32.8	7.87	146.62	0.38
mag	4.44	4.4	0.41	3.5	6.7	1.31
nst	60.97	35.0	76.32	6.0	555.0	3.29
gap	97.27	95.2	46.16	11.0	290.0	0.38
dmin	2.16	1.61	1.45	0.08	8.16	0.84
rms	0.82	0.79	0.23	0.28	1.89	0.65
horizontalError	7.89	7.7	2.35	3.2	15.9	0.55
depthError	7.03	6.7	4.84	1.2	46.9	3.39